

Group Project

CRISPR Screening and Analysis

Online learning week

28 July - 1 August

Group Project

A missense mutation in the coding region of the gene **SRY gene** (**g.2787144A>G, p.C154R**) has recently been found to alter the expression of the protein. This leads to an uncommon disorder in men.

The presence of the mutation does not always lead to disease, suggesting there are other factors at play. Your research team have been tasked with identifying genes involved in disease progression - particularly new drug targets to reverse the disorder.

Groups' tasks



- 1) Describe a CRISPR-based workflow to introduce the mutation in *SRY* in an appropriate cell model and verify the mutation.
- 2) Describe how you would design a CRISPR KO screen to identify modulators of *SRY* expression in this modified cell model.
- 3) Describe next steps after the screen, including a validation strategy.

*For each step, give a **rationale** for your approach and describe the **analysis** in each case.*



Modelling the *SRY* Promoter Mutation - Pointers

- What is your model system?
- Should it naturally express SRY? How about the efficiency of genetic manipulation?
- Your choice of CRISPR tools to introduce a precise genetic alteration
- Optimal method to provide correct copies of the DNA? How would you deliver the tools into the cells?
- What methods will allow you to screen for the correction?
- What methods can be used to measure both RNA and protein levels?

CRISPR Knockout Screen - Pointers



- How will you ensure the mutation is present? What factors will limit your analysis of more genes?
- What readout will allow you to separate cells with higher or lower expression?
- How will you sort and collect the most interesting/informative population of cells?
- What sequencing strategy will allow you to identify sgRNAs?
- How will you leverage the sequencing data to find candidate genes worth investigating further?

Group presentation

15 min per group

10 min for presentation and 5 min for questions